■ PROJECT ECO-BASKET: THE MAGIC FRESHNESS BOX! ■

Look Inside! A Magic Box with 3 Special Rooms and Shiny Traffic Lights! No Plugs, No Wires!

■ CLICK! SLIDE! SEE THE MAGIC DRAWERS OPEN!

The Eco-Basket is fully covered like a big secret box to stop bugs and flies from getting inside! But look closer... pull the handles, and the drawers slide out like a magic trick! Each drawer has its own set of **three little traffic lights** and a tiny timer clock to tell you exactly how fresh your food is!

[■ ROOM 1: THE FRUIT VAULT]
• **Traffic Lights:** ■ ■ ■ Glowing Bright!
• **What it does:** Keeps apples and bananas safe! It blows away their invisible gas balloons so they don't make other fruits sick!

[■ ROOM 2: THE DRY DRAWER]
• **Traffic Lights:** ■ ■ ■ Glowing Bright!
• **What it does:** Keeps potatoes and onions super dry and cozy so they don't grow weird roots or turn mushy!

[■ ROOM 3: THE GREEN SHIELD]
• **Traffic Lights:** ■ ■ ■ Glowing Bright!
• **What it does:** Gives broccoli and leafy greens a fresh bath! It locks in water like a wet blanket so they stay crunchy!

■ WHAT IS IT MADE OF?

• **Smashed Rice Shells (Pressed Rice Husk Chassis):** The strong outer box is made from leftover crunchy shells that grow on rice plants instead of chopping down trees!

• **Shiny Silver Steel (Stainless Steel):** The internal secret rooms are made of clean steel, keeping everything germ-free and easy to wipe clean!

■ WHY IS IT AN AMAZING WIN?

• **Saves Piggy Bank Money:** Food stays fresh longer, so families don't throw away food or waste money!
• **Works Anywhere:** No electricity needed! It works even if the neighborhood lights go completely dark!

PROJECT ECO-BASKET: ENGINEERING LOG & BUSINESS CASE

Target Domain: Off-Grid Urban Interventions & Sustainable Agro-Waste Architectures
Developer Profile: Google Ambassador Program Competitor Submission

1. Technical Engineering & Materials Specification

The Eco-Basket addresses household food decay using thermodynamic principles, functioning as a fully enclosed biological storage module to prevent pest contamination and heat ingress.

Component	Material Choice	Engineering Advantage
Outer Box Chassis	Compressed Rice Husk Waste	High thermal insulation value, zero tree-logging impact, circular agro-economy ir
Internal Drawers	Food-Grade Stainless Steel	Corrosion resistance, non-porous structure preventing rot bacterial anchoring, hy
Vent Flaps	Calibrated Bi-Metallic Strips	Passive autonomous opening via thermal expansion to dump hot humid decomp

2. Operational Dynamics of the Three-Light Freshness Status Module

Each distinct product compartment is outfitted with a mechanical tracking dial paired with an autonomous three-light status module. These indicators actively track real-time food storage windows to prevent cross-contamination and accelerated 'rot chains' without draw on external power grids:

• ■ **Green Light Status (Optimal Freshness):** Indicates ideal atmospheric conditions within the compartment. Produce maintains structural integrity and maximum vitamin retention.

• ■ **Yellow Light Status (Impending Spoilage Warning):** Alerts the household that the internal storage window is narrowing. Immediate cooking is recommended to utilize nutrients and prevent economic waste.

• ■ **Red Light Status (Critical Decay / Isolation Protocol):** Signals advanced structural spoilage. Prompts immediate removal and quarantine of the specific item to prevent volatile gas emitters from degrading neighboring items.

3. Economic Profitability & Daily Life Impact Analysis

For households in dense urban or off-grid environments, monthly grocery spend constitutes a major financial vulnerability. Standard refrigeration remains structurally and financially inaccessible due to unstable electrical infrastructure.

Metric Evaluated	Traditional Storage Baskets	Project Eco-Basket System
Average Spoilage Rate	30% - 35% weekly losses	Reduced to under 8% due to zone segregation
Grid Dependence	High (If using fridge) / N/A	0% Power Required (Completely passive cooling)
Estimated Monthly Savings	0 INR (Constant replacement cost)	Estimated 2,500 - 3,200 INR saved per household
Payback Period (ROI)	Continuous ongoing loss	Fully amortized within 3 months of implementation

4. Project Alignment: Google Ambassador Sustainability Imperative

Project Eco-Basket fulfills the core goals of local civic engineering by shifting reliance away from centralized grid infrastructure. By deploying zero-energy cooling mechanisms coupled with upcycled agricultural processing byproducts, this system builds community resilience, prevents baseline methane generation from municipal landfills, and provides localized economic stabilization.

This is for informational purposes only. For medical advice or diagnosis, consult a professional. AI responses may include mistakes.

PROJECT ECO-BASKET: THE MAGIC FRESHNESS BOX!

HOW IT IS MADE (MATERIALS)

EARTH-LOVING MATERIALS



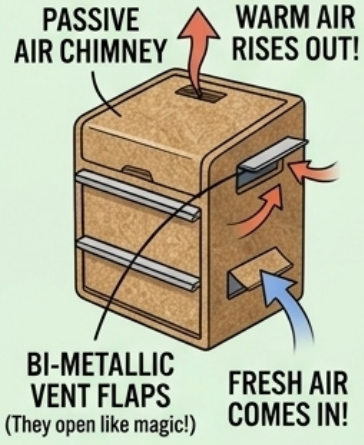
PRESSED RICE HUSK

HYGIENIC STAINLESS STEEL

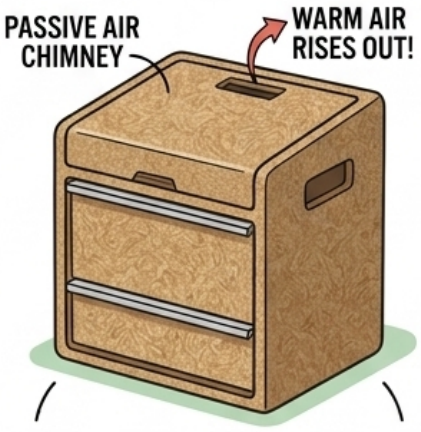
We used leftover rice shells for the big box and strong, clean steel for the secret rooms.

HOW IT WORKS (THE MAGIC SCIENCE)

PASSIVE COOLING POWER



The warm air floats up and out, and fresh air zooms in from the bottom!



1. THE FRUIT VAULT (Gas Emitters)



Keeps bad gases away so other foods don't get sick!

2. THE ROOT DRAWER (Keep Dry!)



Keeps potatoes dry and cozy so they don't grow roots!

3. THE LEAFY GREEN SHIELD (Lock Humidity!)



Locks in water to keep greens super crunchy!

BENEFITS & ADVANTAGES



MONEY SAVER | **NO ELECTRICITY NEEDED!** | **GLOBAL WIN**

Saves mommy & daddy money, works anywhere, and helps the planet by keeping food from going in the trash!

HOW CAN IT BE USEFUL



FRESHNESS EVERYWHERE



WORKS OFF-GRID



CLEAN AND SAFE

Keeps food safe from bugs and works even if the lights go out!

THE COMPARTMENTS OPEN

Slide the drawers like a magic trick to find happy food!

#GOOGLEAMBASSADORPROGRAM | INNOVATION FOR SUSTAINABILITY COMPETITION